

Project 1

TITLE: Smart Environmental Monitor

OBJECTIVE:

To develop an environmental monitoring system using ESP32 and DHT22 sensor for measuring temperature and humidity in real time.

COMPONENTS USED:

- * ESP32
- * DHT22 Sensor
- * Wokwi Simulator

SOFTWARE USED:

- * Arduino IDE
- * Wokwi Simulator

WORKING PRINCIPLE:

The DHT22 sensor measures temperature and humidity values. The ESP32 reads these values periodically and displays them on the Serial Monitor. The system also analyzes environmental conditions and categorizes them as normal, hot, cold, dry, or humid.

FEATURES:

- * Real-time temperature monitoring

- * Real-time humidity monitoring
- * Environmental condition analysis
- * Min/Max value tracking
- * Non-blocking timing using millis()

RESULT:

The system successfully monitored environmental conditions and displayed sensor readings in real time.

CONCLUSION:

This project helped in understanding sensor interfacing, data acquisition, and real-time monitoring using ESP32.